INCH-POUND

MIL-DTL-32106 (OS)
3 March 2003

DETAIL SPECIFICATION

Initiators, Electrical, for Rocket Motor Igniters

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1. Scope. This specification furnishes the general requirements for the manufacture and acceptance of electrical initiators for use in rocket motors.

1.2. Classification. Initiators covered by this specification are categorized as Class A as specified in MIL-I-23659.

1.3. Application. This document covers the requirements for an initiator used to provide an output charge for the ignition of a rocket motor igniter.

2. APPLICABLE DOCUMENTS

2.1. General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents specified in sections 3 and 4 of this specification, whether or not they are listed.

2.2. Government documents.

2.2.1. Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center, Engineering Documentation Branch (Code 4230), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1340

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SPECIFICATIONS

FEDERAL

QQ-A-1876 Aluminum Foil

DEPARTMENT OF DEFENSE

MIL-DTL-16878 Wire, Electrical, Insulated, General Specification for

MIL-I-23659 Initiators, Electric, General Design Specifications for

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-202 Test Method Standard Electronic And Electrical Component Parts

MIL-STD-464 Electromagnetic Environmental Effects Requirements for Systems

MIL-STD-810 Environmental Engineering Considerations And Laboratory Tests

MIL-STD-961 Department of Defense Standard Practice Defense Specifications

MIL-STD-1168 Ammunition Lot Numbering And Ammunition Data Card

(Unless otherwise indicated, copies of Federal and Military specifications and standards are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2.2. Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DOCUMENTS

DOD 4145.26-M DOD Contractors' Safety Manual for Ammunition and Explosives

49 CFR Code of Federal Regulations

(Application for copies should be addressed to Commanding Officer, Naval Surface Warfare Center, Indian Head Division, Documentation Branch (Code 4230), Indian Head, MD 20640-5000.)

2.3. Non-Government publications and drawings. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents, which are DOD adopted, are those listed in the issue of the DODISS cited in solicitation. Unless otherwise

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specified, the issues of documents not listed in DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 1742 Standard Practice for Radiographic Inspection

(Application for copies should be addressed to the American Society for Testing and Materials Customer Service, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

ELECTRONIC INDUSTRIES ALLIANCE (EIA)

J-STD-001 Requirements for Soldered Electrical and Electronic Assemblies

J-STD-006 Requirements for Electronic Grade Solder Alloys and Fluxed and Non-Fluxed Solid Solders for Electronic Soldering Applications

(Application for copies should be addressed to the Electronic Industries Alliance, 2500 Wilson Blvd, Arlington, VA 22201-3834.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

AS3582 Packing, Preformed – O-ring Seal

AS5202 Port or Fitting End, Internal Straight, Thread, Design Standard

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

2.4. Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS Initiators shall be qualified in accordance with section 4 of this specification.

3.1. First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with section 4.2 of this specification.

3.2. Selection of specification and standards. Specifications and standards for necessary commodities and service not specified herein shall be in accordance with MIL-STD-961.

3.3. Materials. Material for the initiators shall be compatible with the explosives, propellants, exhaust composition, pyrotechnic, and ballistic gas. Inspection and testing of materials shall be in accordance with section 4.6 of this specification. The materials shall withstand environmental, electromagnetic, functional, service, and storage conditions to which they may be exposed. Acceptance or approval of materials for design or during the course of manufacture shall in no case be construed as a guarantee of acceptance of

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the finished initiators. Sealant used in close proximity with the energetic materials shall be compatible with these materials. Materials that are nutrients for fungi shall not be used.

3.4. Physical. The initiator shall meet the requirements of Figure 1. Examination of all products shall be made to determine if the initiator meets the requirements with respect to workmanship, marking, conformance to drawings and for any visible defects.

3.5. Mechanical. The initiator shall meet the requirements of Figure 1. Examination of all products shall be made to determine if the initiator meets the requirements with respect to workmanship, marking, conformance to drawings and for any visible defects.

3.6. Electrical. The initiator shall contain either a single or dual bridgewire (depending on the initiator requirements) and meet the requirements of this specification. Refer to Figure 2 for specific details.

3.6.1. Circuit resistance. The initiator circuit resistance shall meet the resistance requirements shown in the Table I measured at 70º ± 10º F. Verification shall be done in accordance with section 4.14.4 of this specification. Failure to meet bridgewire resistance verification per section 4.14.4 of this specification constitutes failure.

TABLE I. Circuit resistance

Initiator Designation	Circuit Resistance (O)
MIL-DTL-32106-1	0.45 – 0.75
MIL-DTL-32106-2	0.45 – 0.60
MIL-DTL-32106-3	0.9 – 1.1*
MIL-DTL-32106-4	0.9 – 1.3
MIL-DTL-32106-5	0.45 – 0.65

*Across pins A-B or C-D

3.6.2. Maximum no-fire stimuli – power current. The initiator shall not fire, dud or otherwise be damaged when subjected to a current of one ampere minimum per bridge with an associated power of one watt minimum per bridge. The initiator shall meet this requirement at 70º ± 5º F. The test method shall be explained in section 4.14.1 of this specification.

3.6.3. Maximum no-fire stimuli – static discharge. . The initiator shall not fire, dud or otherwise be damaged when subjected to the 25,000 volt simulated human electrostatic discharge in accordance with 4.14.2 of this specification.

3.6.4. Maximum no-fire stimuli - stray voltage. The initiator shall be capable of withstanding the effects of a stray voltage environment without pre-igniting (firing). The initiator shall meet this requirement when tested at 70º ± 5º F. The test method shall be as described in section 4.14.3. of this specification.

3.6.5. EMI/RF filter. The unit shall contain an EMI/RF Filter per MIL-STD-464. The initiator filter shall meet the applicable initiator drawing requirements for working voltage, capacitance and amperes when tested at the RF attenuation levels shown in Tables II, IIa and IIb. Verification shall be done in accordance with MIL-STD-464 section 5.6.

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TABLE II. RF attenuation for MIL-DTL-32106-1,-2,-5 initiators

Min Insertion Loss DB	Frequency
15	30KHz
28	150 KHz
33	300 KHz
44	1MHz
60	10 MHz
70	1GHZ

TABLE IIA. RF attenuation for MIL-DTL-32106-3 initiator

Min Insertion Loss DB	Frequency
2	1MHZ
10	10MHZ
12	30MHZ

TABLE IIB. RF attenuation for MIL-DTL-32106-4 initiator

Min Insertion Loss DB	Frequency
3	159 KHz
9	320 KHz
23	1.59 MHz
43	15.9 MHz
63	159 MHz
83	1.59 GHZ

3.7. Functional tests. The initiator shall have the required performance characteristics:

3.7.1. All-Fire. Initiators shall fire under the conditions shown in Table III when tested in accordance with section 4.15.1 of this specification.

3.7.2. Pressure. The initiators shall develop pressures as specified in Table IV when All Fire current is applied.

3.7.3. Leakage. Assembly must withstand 5000 ± 100 psi differential pressure across seal prior to loading output charge without leakage and without permanent thread deformation when tested in accordance with section 4.12 of this specification.

3.7.4. Seal. The output closure disc shall provide a hermetic seal that shall not leak more than $6.102 \times 10^{-8}\ in^3/\text{sec}$ of helium at one atmosphere of differential pressure when tested in accordance with section 4.12.1 of this specification.

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TABLE III. All-fire current requirements

Initiator Designation	Applied Current (Amperes)	Max Applied Pulse (Sec)
MIL-DTL-32106-1	$7.0\pm^{0.50}_{0.00}$	0.020
MIL-DTL-32106-2	$7.0\pm^{0.50}_{0.00}$	0.020
MIL-DTL-32106-3	$3.5\pm^{0.2}_{0.0}$ *	0.020
MIL-DTL-32106-4	$3.5\pm^{0.50}_{0.00}$	0.020
MIL-DTL-32106-5	$7.0\pm^{0.50}_{0.00}$	0.020

* Unit shall fire when specified current and pulse rate is applied to a single bridgewire, A-B or C-D. (See Figure 3)

TABLE IV. Time to reach pressure envelope

Initiator Designation	Pressure (psi)	Closed Volume (in^3)	Time to Reach Max Pressure (Seconds)	Max Pressure (psi)
MIL-DTL-32106-1	1000±150	0.61	0.010	1150
MIL-DTL-32106-2	1450±300	0.61	0.010	1750
MIL-DTL-32106-3	1000±150	0.61	0.010	1150
MIL-DTL-32106-4	1000±150	0.61	0.020	1150
MIL-DTL-32106-5	1000±150	0.61	0.010	1150

3.8. Shock. The initiators shall be capable of withstanding shock conditions as specified in section 4.9 of this specification. After being subject to the shock test, the initiators shall meet the design performance requirements when test fired.

3.9. Vibration. The initiators shall be capable of withstanding vibration conditions as specified in section 4.10 of this specification. After being subject to the vibration test, the initiator shall meet the design performance requirements when test fired.

3.10. Temperature, humidity, altitude cycling (THA). The initiators shall be capable of withstanding temperature – shock/humidity/altitude conditions as specified in section 4.11. of this specification and shall meet the design performance requirements when test fired.

3.11. Visual Inspection. Visual inspection shall be in accordance with section 4.4 of this specification.

3.12. Radiographic examination. Radiographic examination shall be performed on all initiators in accordance with ASTM E 1742. The radiographic inspection shall be conducted to verify the presence and proper orientation of components. The radiographic inspection shall also verify that there are no foreign materials present, no missing or improperly positioned assemblies, no cracks, no voids or fissures in charges. The radiographic inspection shall also verify that the pyrotechnic charge is consistent in height and density with respect to each other and bridgewires are in contact to initiator pins. The radiographic

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inspection shall also verify there are no gaps between the bridgewire and the explosive charge. Radiographic examination shall be done in accordance with section 4.7 of this specification.

3.13. Workmanship. Initiators and components shall be free from foreign material, dents, scratches, split or components, poor quality welds, chipping or flaking of painted or plated metal surfaces, sharp edges or other defects or discontinuities as specified on applicable drawing.

3.14. Soldering. Soldering shall be in accordance with J-STD-001, using solder in accordance with J-STD-006, SN 63 WRAP 2.

3.15. Labels and markings. The initiator shall be marked in accordance with MIL-R-81128, paragraph 3.3.2.

3.16. Packaging. All initiators shall be shorted and be completely enclosed in aluminum foil meeting the requirements of QQ-A-1876. Initiator MIL-DTL-32106-3 shall have its contact pins, A, B, C, and D shorted and the unit shall be completely enclosed in Aluminum foil. The packaging of initiators MIL-DTL-32106-1, -3, -4, and –5 shall be in accordance with federal regulation code 49 CFR part 173 in general.

3.17. Lot numbering. Lot numbering shall be in accordance with MIL-STD-1168.

4. VERIFICATION Qualification testing of initiator design shall be performed in accordance with MIL-I 23659, General Design Specifications For Initiators, section 4.3.

4.1 Classification of inspections. The inspection requirements specified herein are as follows:

a. First article inspection (see 4.2)
b. Quality conformance inspection (see 4.3)

4.1.1 Certification of supplier. The vendor shall meet all the requirements shown in Table V to certify as a producer of the initiators.

TABLE V. Certification

The vendor shall meet all the requirements of this specification.
The vendor shall meet the requirements of this specification.
The vendor shall produce an initiator with an Explosive Hazard Classification 1.4C.

4.2 First article inspection. Inspection shall be performed after award of contract and prior to production by a facility designated in the contract order. First article inspection shall be performed on sample units, which have been produced using the same materials, equipment, processes and procedures as will be used in production. An acceptable first article sample of fifty-seven (57) initiators shall be prepared using the same procedures and methods proposed for the production lot. Three (3) of those initiators will be used for investigative purposes. All parts and materials, including packaging and packing shall be the same as used for production and shall be from the intended sources of supply. First article test shall be in accordance with Table VI. Failure of any initiator shall be a cause for first article rejection.

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4.3 Quality Conformance Inspection.

4.3.1 Lot Definition. The lot shall consist of those units required by the Customer's Purchase Order, including those units required for destructive testing. The lot shall be fabricated from the same lot of materials and assembled using the same manufacturing processes and procedures. Explosives shall be from the same batch. Lots shall be sequentially numbered in the order in which they are produced.

TABLE VI. First article test matrix (1 watt/1 amp)

Tests	Ref. Para.	No. Of Initiators (Groups) & Test Sequence (A, B, C…)						Total
		I	II	III	IV	V	VI	
Quantity of Initiators		6*	6*	6*	12*	12*	12*	54
Dimensional Inspection	4.13	A	A	A	A	A	A	54
EMI/RF	4.14.5	B**	B **	B **	B **	B **	B **	54
Visual	4.4	C	C	C	C	C	C	54
Radiographic	4.7	D	D	D	D	D	D	54
Seal	4.12.1	E	E	E	E	E	E	54
Resistance	4.14.4	F	F	F	F	F	F	54
Electrostatic Discharge	4.14.2	G	G	G	G	G	G	54
Resistance	4.14.4	H	H	H	H	H	H	54
Stray Voltage	4.14.3	I						6
Resistance	4.14.4	J						6
Shock	4.9		I					6
Vibration	4.10			I				6
Temperature Humidity Altitude	4.11				I			6
Radiographic	4.7		J	J	J			24
Leakage	4.12		K	K	K			24
Electrostatic Discharge	4.14.2		L	L	L			24
Power Current	4.14.1	K	M	M	M			30
Resistance	4.14.4	L	N	N	N	I		42
All-Fire	4.15.1	M	O	O	O	J	I	54

*The quantity listed in the column identifies the quantity of initiators within each numbered group.
**The total number of initiators (54) shall be EMI/RF tested prior to being loaded with output charge.

4.3.2 Non-destructive lot acceptance test. Each initiator shall be subjected to the inspections outlined in Table VII, Lot acceptance test (LAT) for non-destructive inspection. Failure of any sample initiator to comply with the requirements listed in Table VII shall be cause for rejection of the sample initiator.

4.3.3 Destructive lot acceptance test (DLAT). Tests required shall be conducted as shown in Table VII. The DLAT sample shall be randomly selected from those units having successfully completed the requirements of non-destructive LAT per 4.3.2. A lot shall be rejected when an initiator fails to function on the first firing attempt of the DLAT (see Table VIII).

4.3.4 Sample size lot acceptance tests. Sampling shall be in accordance with Table IX.

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TABLE VII. Lot acceptance test (non-destructive)

Inspection (100%)	Requirement Paragraph	Test Paragraph
Dimensional Inspection	3.4	4.13
Visual Examination	3.11	4.4
Bridgewire Resistance	3.6.1	4.14.4
Radiographic Inspection	3.12	4.7
Seal	3.7.4	4.12.1

TABLE VIII. Destructive lot acceptance test (DLAT)

Test (sample size)	Requirement Paragraph	Test Paragraph
No Fire	3.6.2	4.14.1
Bridgewire Resistance	3.6.1	4.14.4
Functional Test	3.7	4.15
Leakage	3.7.3	4.12

TABLE IX. Lot conformance sampling size

Lot Size	Test Sample Size	Retained Sample Size
2 to 25	3	3
26 to 50	8	3
51 to 90	13	3
91 to 150	20	3
151 to 280	32	3
280 to 500	50	3
501 to 1200	75	3
1201 to 3200	100	3
3201 to 10,000	150	3

4.4 Visual inspections. The external condition and appearance of the initiators shall be determined by comparison to drawings as required. All initiator assemblies shall be visually examined and compared to the appropriate drawing for proper assembly, identification, color and interface control points as per figure 1. Those with illegible or inadequate descriptive markings or serial numbers, dents, deep scratches, or any other defects, which may prevent proper installation or functioning, shall be rejected. Any visual defects found during nondestructive inspection and destructive tests will result in 100 percent screening of the production lot at the contractor's expense. Each initiator shall meet the requirements for dimensions and markings. Deviations from the control drawings may be cause for rejection of the sample.

4.5 Production lot. Production lots shall be in accordance with MIL-STD-1168. Initiators required for all test purposes are randomly selected from the production lot and will not apply as part of the quantity specified for delivery by the contract. Explosives from a single lot shall be used in a production lot of initiators. Explosives from one production lot may be used in more than one initiator production lot.

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4.6 Materials. Inspection and testing of component parts and assemblies shall be made to determine compliance with section 3.3 of this specification. Where defects, incompatibility or inferior quality is evident and the Government design agency deems a material analysis, the contractor will be required to submit samples or specimens to the contracting officer for analysis and approval.

4.7 Radiographic examination. X-ray radiographic examination shall be conducted in accordance with ASTM E 1742, Level 2-2T minimum. X-ray shall be performed in 1 view perpendicular to the longitudinal axis of the initiator. All initiators shall be identified with serial numbers beginning with 001 prior to examination. The initiators shall be arranged on boards or trays in consecutive order, with any missing serial numbers identified on the radiographic plate. Each radiograph shall carry a permanent identification of the items displayed thereon in a 4x6-inch region maximum. The radiographic identification shall include the drawing number, the complete lot number in accordance with MIL-STD-1168, the contract number, and the span of serial numbers displayed. Radiographs of the entire production lot shall accompany the lot acceptance test results. Any observable imperfections shall be cause for rejection of the initiator. Defective initiators found during radiographic review are to be marked on the radiographic plate and removed from the production lot. Settings on the radiographic equipment shall be noted on the radiographic review sheets.

4.8 Examination of product. Examination of all products shall be made to determine if the initiator meets the requirements with respect to workmanship, marking, and conformance to drawings and for any visible defects.

4.9 Shock. To determine if the initiators meet the shock requirements of section 3.8 of this specification, the initiators shall be tested as follows. Initiators are to be mounted in the device of intended use or a suitable test vehicle.

If the test is conducted using a suitable test vehicle, the degree of support shall be the same as that offered by the device of intended application. The shock pulse shall be applied to the initiator's mounting points in both directions along each of two mutually perpendicular axes. The shape of each shock pulse shall approximate as nearly as possible a half sine wave. The amplitude of each shock pulse shall exceed 200 g's for 1.5 ± 0.4 milliseconds and it shall exceed 65 g's for 9 ± 0.9 milliseconds. Initiators shall be free from visible damage or leaks (if applicable), and shall perform satisfactorily in function tests subsequent to this test.

4.10 Vibration. To determine if the initiators meet the requirements of section 3.9 of this specification, they shall be tested to the vibration test of table 514-1 aircraft category, procedure 1, parts 1, 2 and 3, with curve H of MIL-STD-810 except that each resonant and cycling period shall be divided equally among –65º F., 70º F. and 200º F. The time of vibration shall be continuous and not accumulative at each temperature.

4.11 Temperature, humidity, altitude cycling (THA). To determine if the initiators meet the requirements of section 3.10 of this specification, the initiators shall be subjected to the temperature-shock/humidity/altitude test as outlined below.

The schedule has been arranged in such a manner that operations are not required outside normal working hours except for such supervision as may be necessary to insure proper operation of he test equipment.

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It is not mandatory that the day/clock time schedule given below be followed: however, it is mandatory that the time, environmental, and sequence requirements be adhered to. It is also noted that only two conditioning chambers are required to accomplish this test. No less than two chambers may be used and still accomplish the temperature shock portion of this test. If it is desired to use three chambers, it is permissible provided the time, environmental, and sequence requirements are met. The fluctuations from the specified temperatures shall not exceed 5º F. When the temperature/altitude chamber door is opened to place test items inside, the chamber pressure will become atmospheric. The time required to return the chamber pressure to 0.65 pounds per square inch (psi) shall not exceed 1 hour.

Test specimens shall be supported on screen trays or racks so that all areas are exposed to the prescribed atmospheric conditions at all times throughout the test.

Procedure

Monday	0800 – Place test items in a chamber maintained at +70° F at 50 percent relative humidity (RH).
	1200 – Raise chamber temperature to +160° F and the RH to 95 percent. the chamber temperature shall reach +160° F at 95 percent RH not later than 1300.
	1600 – Remove test items from above chamber and immediately place in a chamber maintained at –65° F at a pressure altitude of 70,000 feet. (0.65 psi)
Tuesday	0800 – Remove test items from above chamber and immediately place in a chamber maintained at +70° F at 50 percent RH.
	1200 – Remove test items from above chamber and immediately place in a chamber maintained at -65° F at a pressure altitude of 70,000 feet.
	1600 – Remove test items from above chamber and immediately place in a chamber maintained at +160° F at 95 percent RH.
Wednesday	0800 – Reduce chamber temperature to +70° F at 50 percent RH. The chamber temperature shall reach +70° F at 50 percent RH not later than 0900.
	1200 – Raise chamber temperature to +160° F at 95 percent RH. The chamber temperature shall reach +160° F at 95 percent RH not later than 1300.
	1600 – Remove test items from above chamber and immediately place in a chamber maintained at –65° F at a pressure altitude of 70,000 feet.
Thursday	0800 – Remove test items fro above chamber and immediately Place in a chamber maintained at +70° F at 50 percent RH.

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1200 – Remove test items from above chamber and immediately place in a chamber maintained at -65° F at a pressure altitude of 70,000 feet. (0.65 psi)

1600 – Remove test items from above chamber and immediately place in a chamber maintained at +160° F at 95 percent RH.

Friday 0800 – Reduce chamber temperature to +70° F at 50 percent RH. The Chamber temperature shall reach +70° F at 50 percent RH not Later than 0900.

1200 – Raise chamber temperature to +160° F at 95 percent RH. The chamber temperature shall reach +160° F at 95 percent RH not later than 1300.

1600 – Remove test items from above chamber and immediately place in in a chamber maintained at -65° F at standard ambient pressure.

This schedule shall be followed for a total of 4 weeks (28 days) except that on the second and fourth weekends the soak time shall be from 1200 on Friday until 0800 on Monday at a temperature of +160° F at 95 percent RH. At the conclusion of the temperature-shock/humidity/altitude test, the items shall be allocated to the tests specified in Table I.

4.12 Leakage. When tested in a AS5202-03 Boss at +70 °F ± 10 °F, the initiator shall be able to meet the requirements of section 3.7.3.

4.12.1 Seal. The output closure disc shall provide a hermetic seal that shall not leak more than 6.102×10^{-8} in^3/sec of helium at one atmosphere of differential pressure. Initiators that do not meet the seal requirement shall be cause for rejection of the sample initiator.

4.13 Dimensional inspection. Initiator shall be 100 percent dimensionally inspected for conformance to the applicable drawing.

4.14 Electrical. Each initiator shall meet the requirements of section 3, electrical tests and pass radiographic inspection.

4.14.1 Maximum no-fire stimulus – power current. To determine if the initiator meets the power current stimulus of section 3.6.2, a direct current of not less than one ampere supplying a minimum of one watt shall be applied to the bridge circuit for a period of at least five minutes. For an initiator having more than one bridge, the current shall be applied to all bridge circuits. The test current shall be regulated throughout the period of application to within 2 percent. If a rectified current is used, the ripple content shall not exceed 5 percent rms of the test current. The initiator shall be conditioned at 70º ± 5º F. The initiators shall be tested for conformance to no-fire power current requirements of section 3 as specified in the verification of MIL-I-23659, section 4.4.3, maximum no-fire stimuli, class A, power current. Initiators that do not meet the requirements of paragraph 3.6.2 shall be cause for rejection of the first article test lot.

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4.14.2 Maximum no-fire stimulus – static discharge. To determine if the initiator meets the requirements of section 3.6.3, a 500 ± 5 percent picofarad capacitor charged to 25000 ± 500 volts and 5000 ± 5 percent ohm resistor shall be connected in a 5 microhenry total inductance series circuit between pairs of pins or leads in all combinations and between the shorted pins or leads (all pins or leads shorted to each other external to the initiator) and the case of the initiator. The series connection shall be maintained for 60 seconds. Switching in this circuit shall be accomplished by bringing together two 0.5 ± 0.05 inch spherical metal electrodes from an initial separation of 3 inches in air. Each series test shall constitute a separate test. Initiators used for this test shall be temperature conditioned for a minimum time of 12 hours at 70º ± 5º F.

4.14.3 Maximum no-fire stimulus – stray voltage. To determine if the initiator meets the requirements of section 3.6.4 of this specification, each initiator shall be subjected to 2000 pulses of direct current. Each pulse shall be of 300 milliseconds duration and pulse rate shall be 2 per second. Each pulse shall have a minimum amplitude of 100 ± 5 milliamps. The initiator shall be temperature conditioned at 70º ± 5º F for a period of 12 hours.

4.14.4 Bridgewire resistance. To determine if the initiator meets the requirements of section 3.6.1 of this specification the test method shall be as follows: For an initiator which does not contain a bridge circuit gap the resistance of each bridge circuit shall be measured with an accuracy of one percent using a test which subjects the bridge circuit to a current of less than 50 milliamperes. These tests shall be conducted in accordance with MIL-STD-202, method 303 except that the measurement shall be made at or corrected to 70º F.

4.14.5 EMI/RF filter. The contractor's test method for verification of the filter for safety against Hazards of Electromagnetic Radiation to Ordnance (HERO) shall be submitted to the Government prior to testing.

4.15 Functional test

4.15.1 All-fire. The initiators designated MIL-DTL-32106-1, -2, -4, -5 shall fire when current of 7.0 – 7.5 amperes with a 20 milli-second max pulse is applied to the positive lead wire at 70° ± 10° F while the negative lead wire is grounded. Failure of bridgewire to ignite the initiator shall be cause for rejection.

The initiator designated MIL-DTL-32106-3 shall fire when current of 3.5 - 3.7 amps with a 20 milli-second pulse max is applied to a single bridgewire A-B or C-D (See Figure 3) at 70° ± 10° F. Failure of bridgewire to ignite the initiator shall be cause for rejection.

4.15.2 Pressure. Initiators shall meet the requirements of section 3.7.2 of this specification. The unit shall develop the required minimum pressure when all-fire current is applied to the initiator. The initiators shall develop the required minimum pressure within the time specified in Table IV of this specification. The test chamber must be approved by the procuring Government agency.

4.15.3 Misfire. An initiator that does not fire under the conditions specified shall be considered defective and the lot considered unacceptable.

4.15.4 Retest. There shall be no retest.

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4.15.5 Test failure. If the test failure is attributable to an assignable cause, excluding the test initiator, the original test results shall be discarded and the test re-conducted.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department or Defense Agency automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1.Intended Use. These initiators are intended for use in military rocket motors. The initiators are designed for this use only, and thus have no commercial application.

6.2.Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1, 2.2.2, and 2.3).
c. Whether first article inspection is required and, if so, specify the test activity (see 3.1 and 4.2).
d. Production lot size and test activity (see 4.3.2. - 4.3.4.)
e. Items of data required for each first article and production lot (see 4.5)
f. Packaging and marking (see 5.1 and 6.4)
g. That the safety precaution requirements of the "DOD Contractors' Safety Manual for Ammunition and Explosives," DOD 4145.26-M, are applicable and should be specified in the contract as required by Federal Acquisition Regulation (FAR) 23.3. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity the safety precaution requirements of "Ammunition and Explosives Ashore," OP 5, are applicable.
h. Data Requirements. Reports of all first article inspection procedures, First article inspection reports, Production acceptance inspection procedures, Production inspection reports and Ammunition Data Cards should be submitted for each lot.

6.3 Data. For the information of contractors and contracting officers, any of the data specified in (a) subparagraphs below, (b) applicable documents listed in section 2 of this specification or (c) referenced lower-tier documents need not be prepared for the Government unless specified in the contract or order.

6.3.1 Standard Deliverables. In addition to the delivery of initiators, the contract may be required to deliver the following:

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a. Product baseline list
b. Radiographic plates
c. Radiographic review report
d. Radiographic sample plates and techniques
e. Requests for waiver/deviation and engineering change proposals (ECP)
f. Progress and management report
g. Test and inspection report if contractor tested
h. Ammunition data cards

6.4 Markings. Markings of exterior containers should be in accordance with Regulation 49 CFR 171-179. The contractor or his agent should apply markings.

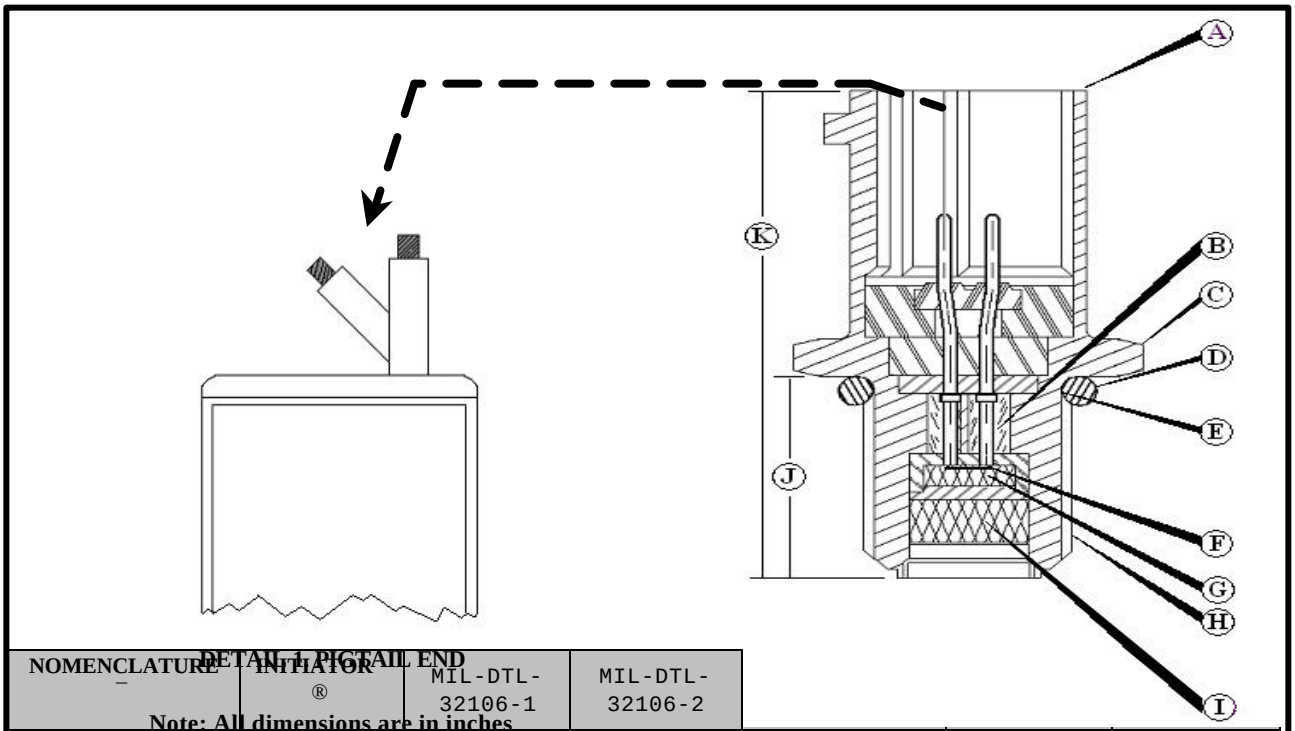
6.5 Subject term (keyword) listing.

JATO

Preparing activity:
Navy – OS
(Project 1340-1030)

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FIGURE 1. Drawing for (1 watt)/(1 amp) class initiator



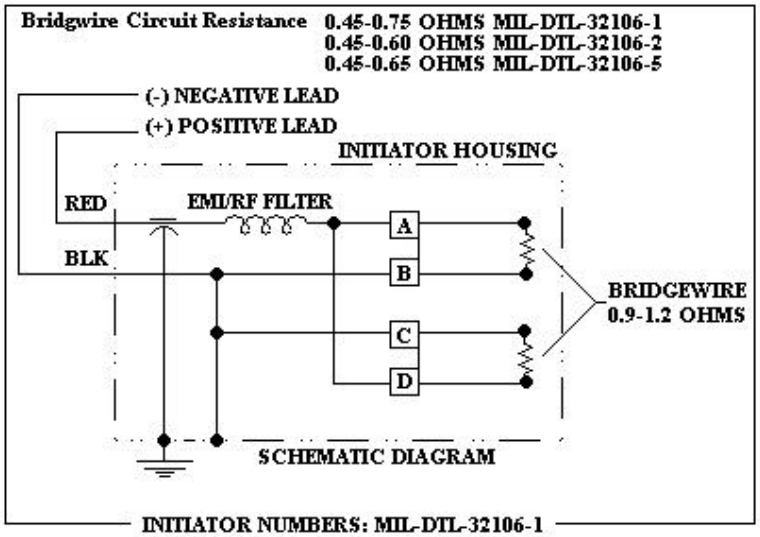
DETAIL 1 PIGTAIL END

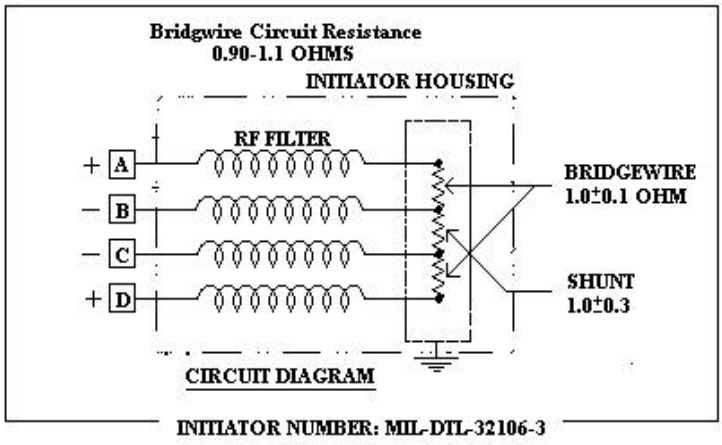
Note: All dimensions are in inches

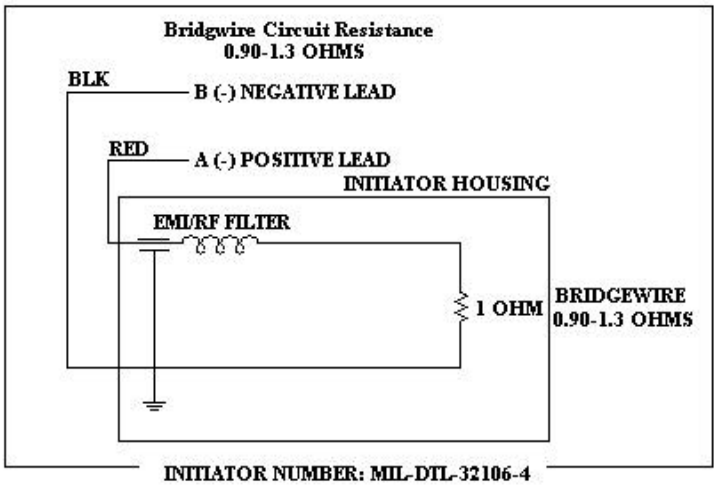
NOMENCLATURE – INITIATOR ®		MIL-DTL-32106-1	MIL-DTL-32106-2			
CONNECTOR END	A	2-AWG 22 PIGTAILS, 61 INCHES LONG (SEE DETAIL 1)	2-AWG 22 PIGTAILS, 10 INCHES LONG (SEE DETAIL 1)	MATES WITH MS27467T11F04S CONNECTOR (4PIN)	2-AWG 22 PIGTAILS, 61 INCHES LONG (SEE DETAIL 1)	2-AWG 22 PIGTAILS, 12 INCHES LONG (SEE DETAIL 1)
GLASS SEAL	B	LENGTH OF GLASS = .20" WEB (PLCS.)	LENGTH OF GLASS = .195" WEB (4 PLCS.)	LENGTH OF GLASS = .20" WEB (PLCS.)	LENGTH OF GLASS = .20" WEB (PLCS.)	LENGTH OF GLASS = .20" WEB (PLCS.)
BODY	C	Ø +540/500 HEX MAT'L= CRES, 304L	Ø +540/500 HEX MAT'L= CRES, 304L	.810 HEX MAT'L; CRES, 304L	Ø +540/500 HEX MAT'L= CRES, 304L	Ø +540/500 HEX MAT'L= CRES, 304L
O-RING	D	AS 3582-009	AS 3582-009	3-306	AS 3582-009	AS 3582-009
THREAD RELIEF	E	.309-.314 DIAMETER	.309-.314 DIAMETER	.478-.483 DIAMETER	.309-.314 DIAMETER	.309-.314 DIAMETER
BRIDGEWIRE COMPOSITION	F	65mg – 500488 ($ZrKClO_4$)	65mg – 500488 ($ZrKClO_4$)	65mg – 500488 ($ZrKClO_4$)	65mg – 500488 ($ZrKClO_4$)	65mg – 500488 ($ZrKClO_4$)
BRIDGEWIRE	G	Material: .002 DIAMETER 'NILSTAIN'	Material: .002 DIAMETER 'NILSTAIN'	Material: .002 DIAMETER 'NILSTAIN'	Material: .002 DIAMETER 'NILSTAIN'	Material: .002 DIAMETER 'NILSTAIN'
THREAD TYPE	H	3/8-24 UNF-3A x .33THD	3/8-24 UNF-3A x .33THD	9/16-18UNF-3A x .556TD	3/8-24UNF 3Ax0.33	3/8-24 UNF-3A x .33THD
OUTPUT CHARGE	I	220mg $BKNO_3$ (1000 psi in $0.61 in^3$ VOLUME)	Titanium Hydride Perchlorate 200mg (1,450 psi in $0.61 in^3$ VOLUME)	220mg $BKNO_3$ (1000 psi in $0.61 in^3$ VOLUME)	220mg $BKNO_3$ (1000 psi in $0.61 in^3$ VOLUME)	220mg $BKNO_3$ (1000 psi in $0.61 in^3$ VOLUME)
MAX PRESSURE	N/A	1150 psi	1750 psi	1150 psi	1150 psi	1150 psi
Dimension I	J	.330 MAX	.330 MAX	.566 MAX	.330 MAX	.330 MAX
Dimension II	K	1.150 MAX	1.150 MAX	1.425 MAX	1.315 MAX	1.150 MAX

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FIGURE 2. Initiator circuit diagram

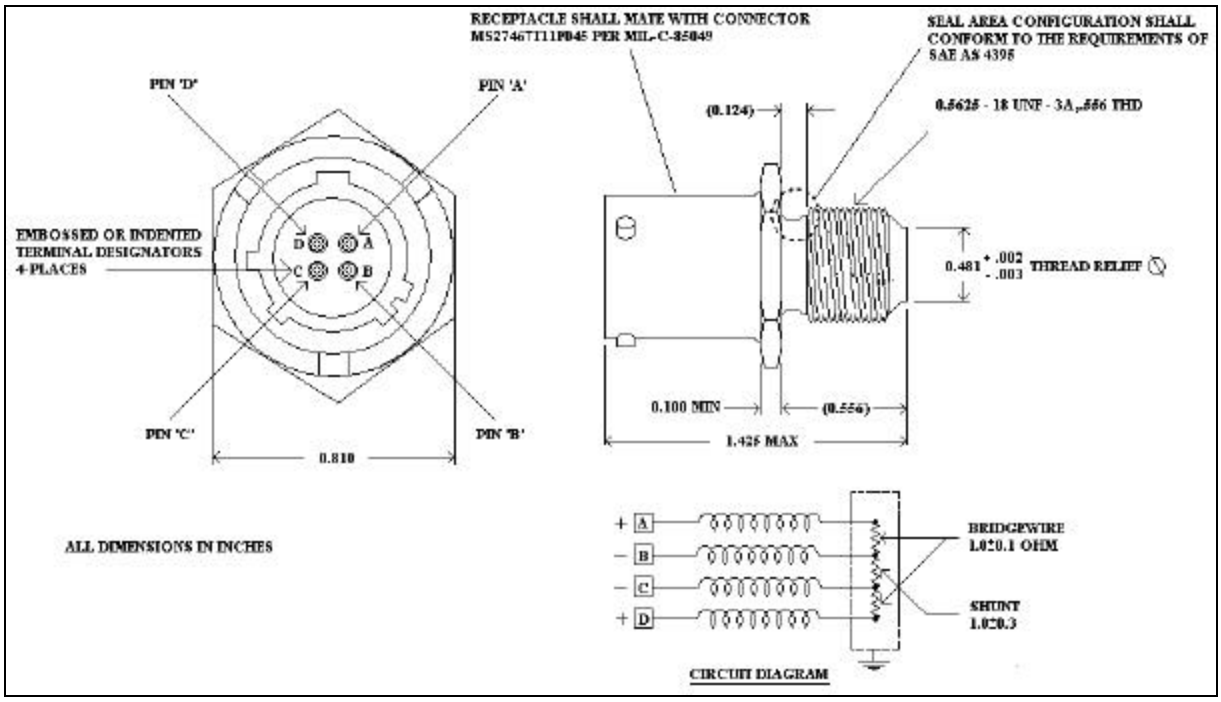






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FIGURE 3. MIL-DTL-32106-3 Receptacle Diagram



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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-DTL-32106	2. DOCUMENT DATE (YYMMDD) 030303

3. DOCUMENT TITLE
INITIATORS, ELECTRICAL, FOR ROCKET MOTOR IGNITERS

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
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8. PREPARING ACTIVITY

a. NAME Commander, Indian Head Division Naval Surface Warfare Center	b. TELEPHONE (*Include Area Code*) (1) Commercial 301-744-1973 (2) DSN 354-1973
c. ADDRESS (*Include Zip Code*) Engineering Documentation Branch (Code 4230) 101 Strauss Avenue Indian Head, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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